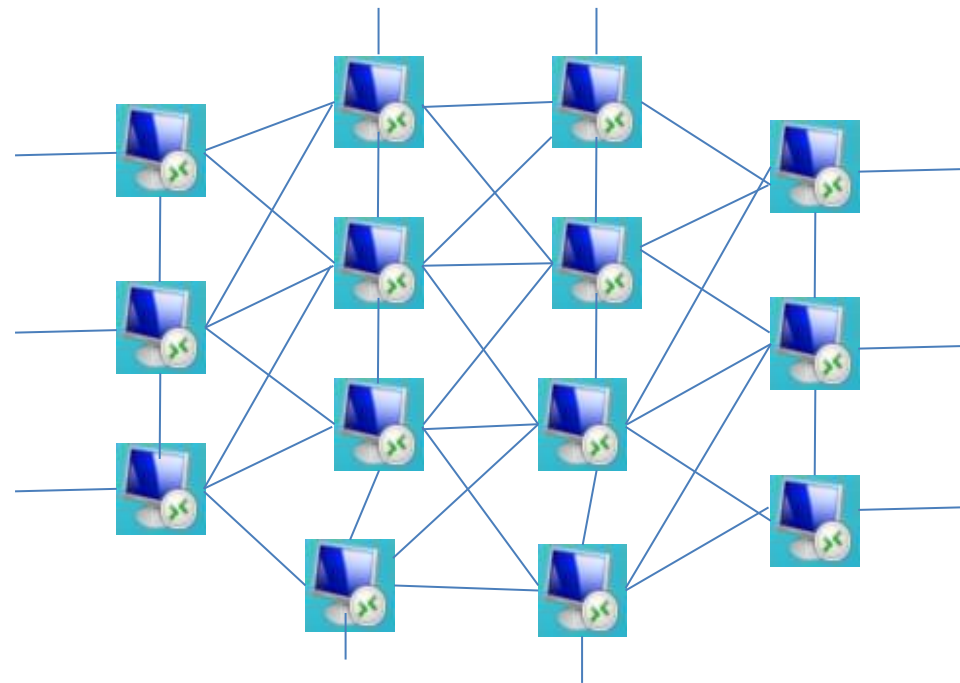




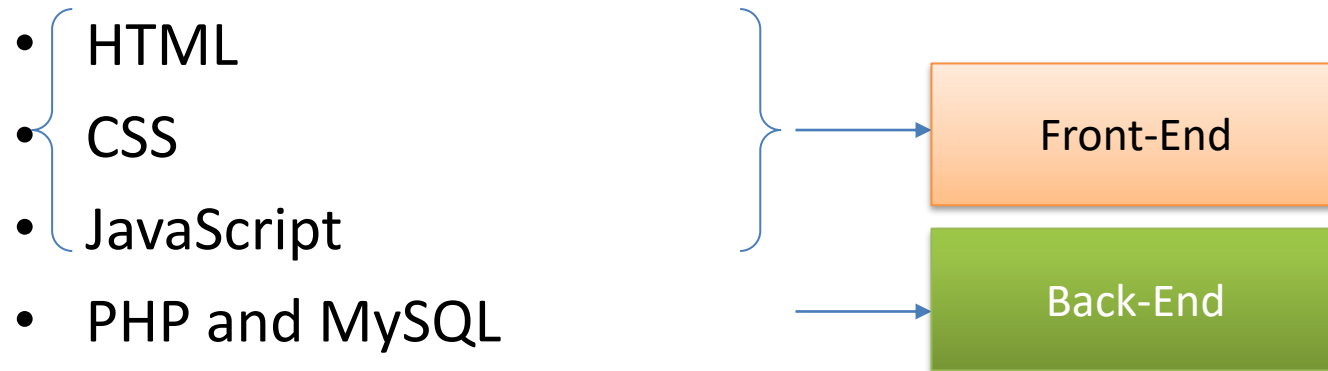
WEB TECHNOLOGIES



Mr. Eric HITIMANA



- Introduction to Internet and Web



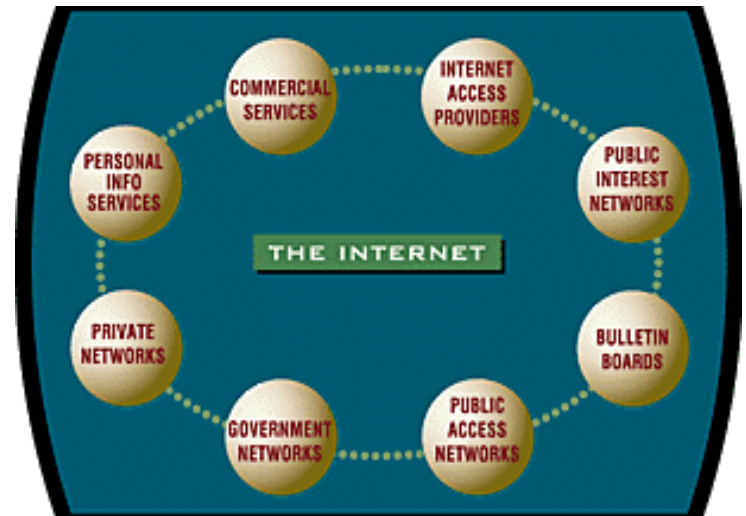


-
- Introduction to the Internet
 - The use of Internet
 - Internet Properties
 - History and Overview of the Web
 - Overview on Web Development



1. Introduction

- What is Internet?
 - The internet has been called the NETWORK of NETWORKs.
 - It is a way to link millions of people around the world.
 - It is the **global system** of Interconnected networks of computers, with different types of computers.





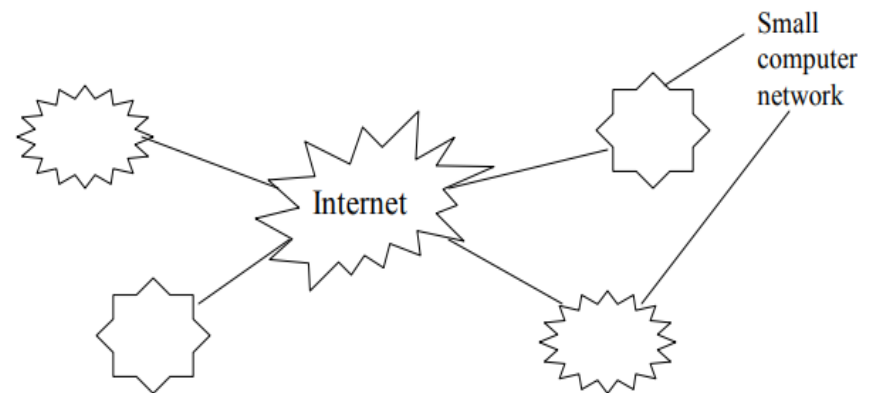
2. The Use of Internet

- Internet is today one of the most important part of our daily life.
- The best uses of the Internet.
 - Communication
 - Research
 - Education
 - Financial Transactions
 - Real Time Meteo-logical Data Updates to Community
 - Online Booking
 - Job search
 - Shopping



3. How does Internet work

- What makes the Internet Works?
 - Internet **protocols** are set of rules that govern how computers talks over the net.
 - The internet properties
 - Interoperable
 - Global
 - Cost is shared
 - Heterogeneity





4. How does Web work

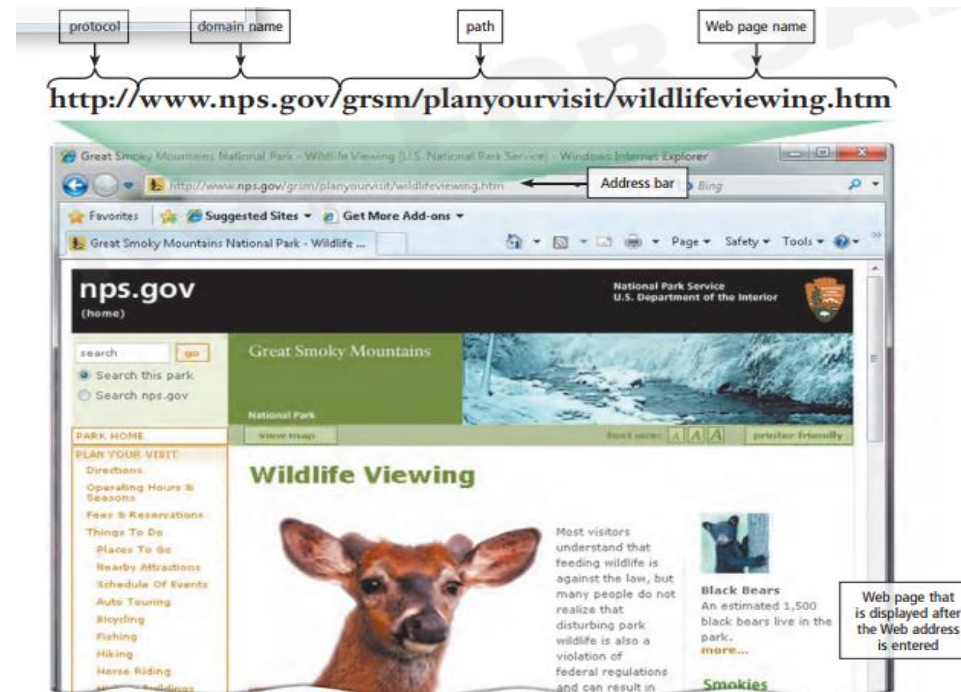
- Internet Addresses?
 - The Internet relies on an addressing system like the [postal service](#) to send data and information to a specific destination.
 - An **IP address**, is used to uniquely identify the devices.
 - A **domain name** is the text version of an IP address.





4. How does Web work

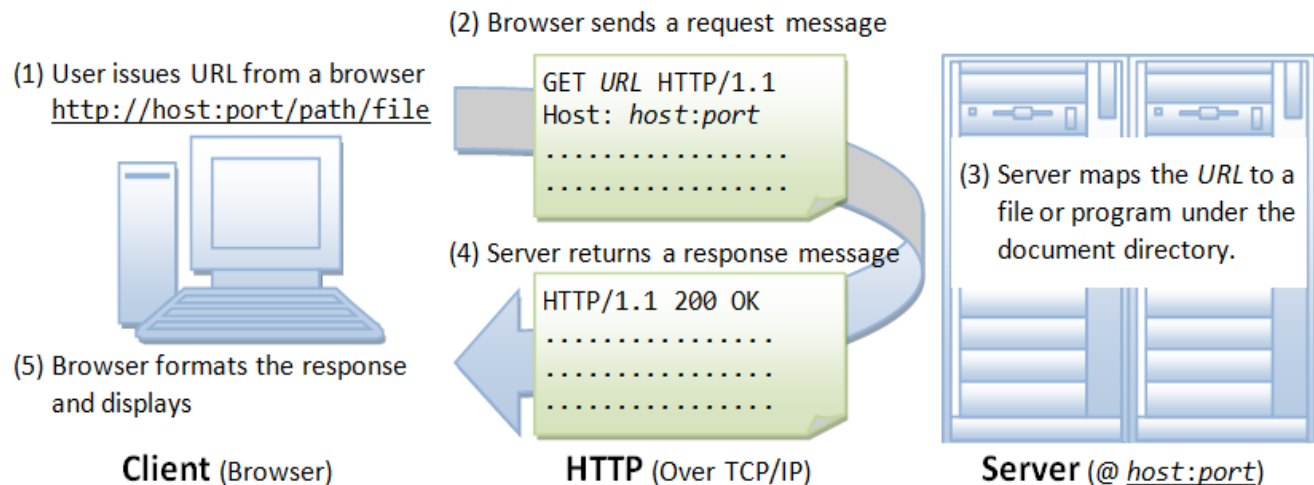
- Web Addresses?
 - The web page has a unique address, called **URL**- Uniform Resource Locator or web address.
 - **Protocol: *//hostname: port/path-and-file-name***





4. How does Web work

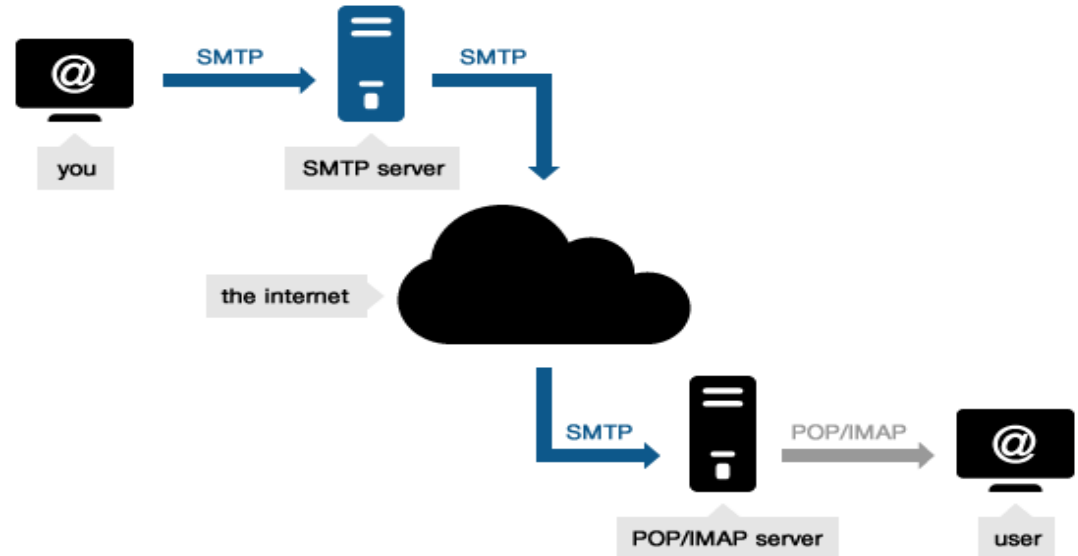
- Protocols governing the Web?
 - **Hyper Text Transfer Protocol (HTTP) -1.**
 - ❖ An asymmetric request-response client-server protocol





4. How does Web work

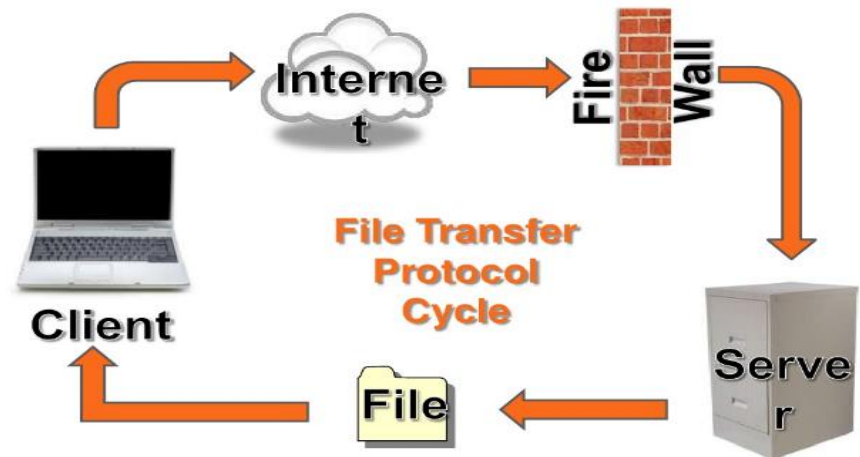
- Protocols governing the Web?
 - **Simple Mail Transfer Protocol (SMTP)-2.**
 - ❖ Used for transferring e-mail between computers. Handle emails exchange and delivery across IPs. It works with POP3, IMAP.





4. How does Web work

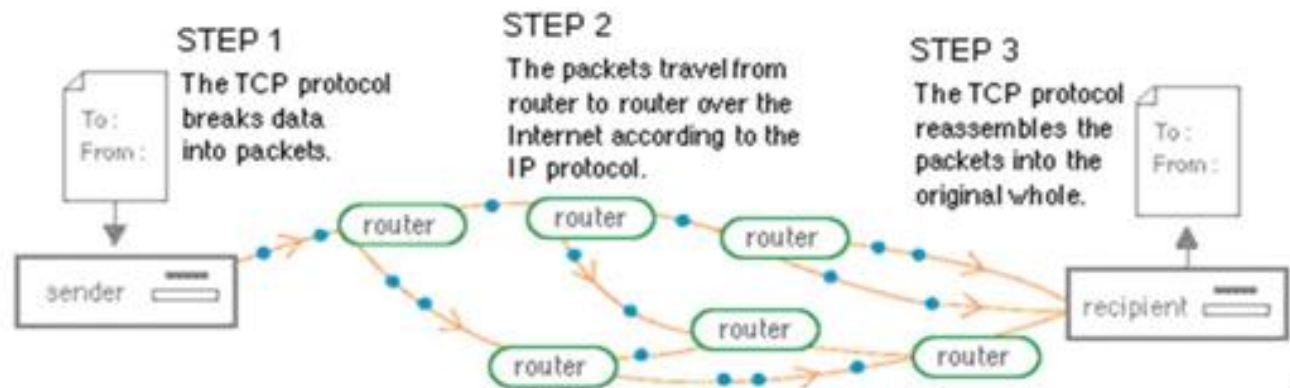
- Protocols governing the Web?
 - **File Transfer Protocol (FTP)-3.**
 - ❖ Used to upload files on server and download files from the server.
 - ❖ **Steps or States:** Request, Response, Transfer, Terminate.





4. How does Web work

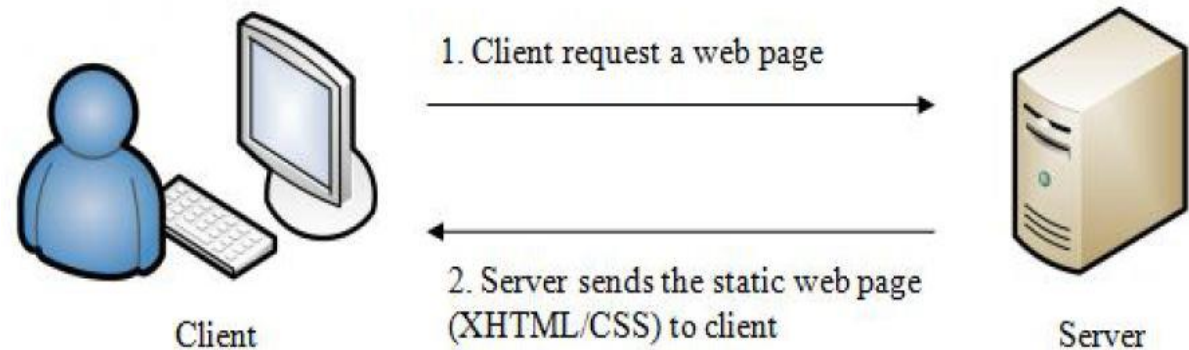
- Protocols governing the Web?
 - **Transmission Control Protocol/Internet Protocol (TCP/IP)-4.**
 - ❖ Is a communication protocol to connect hosts to the Internet.
 - ❖ **Steps 1:** The TCP protocol breaks data into packets.
 - ❖ **Steps 2:** The packets travel from router to router over the internet according to the IP protocol
 - ❖ **Step 3:** The TCP protocol reassembles the packets into the original whole.





5. Introduction to Web Design

- Two types of Web sites
 - **Static web sites:**
 - Has no friendly updatability.
 - Static data

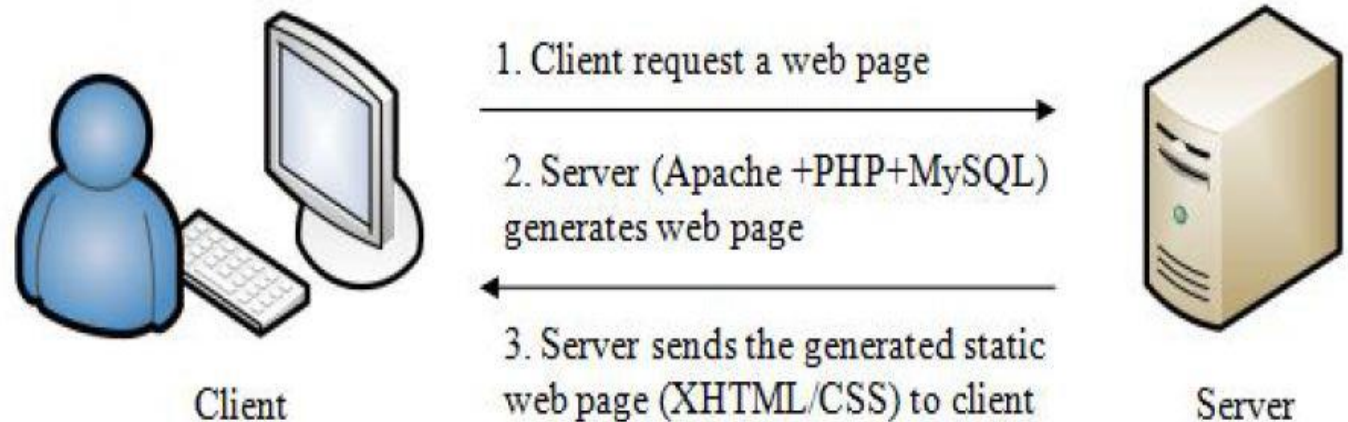




5. Overview on Web Development

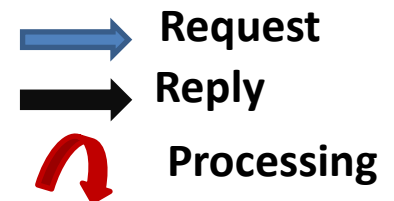
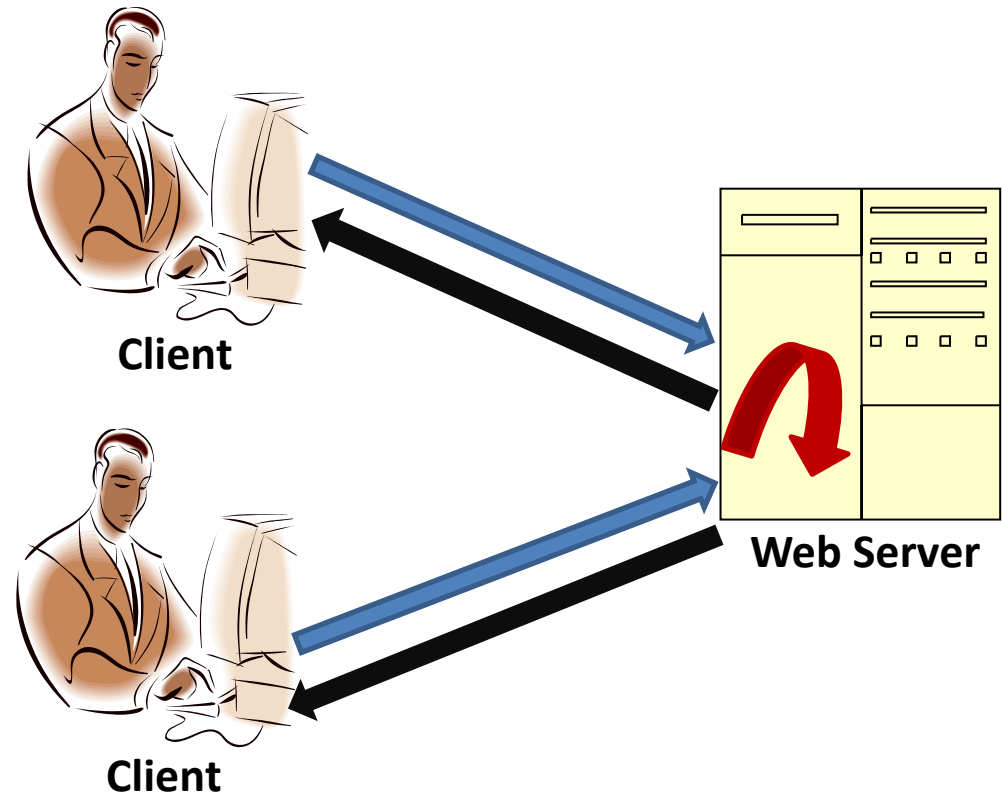
– Dynamic web sites:

- Friendly updatability.
- Dynamic data





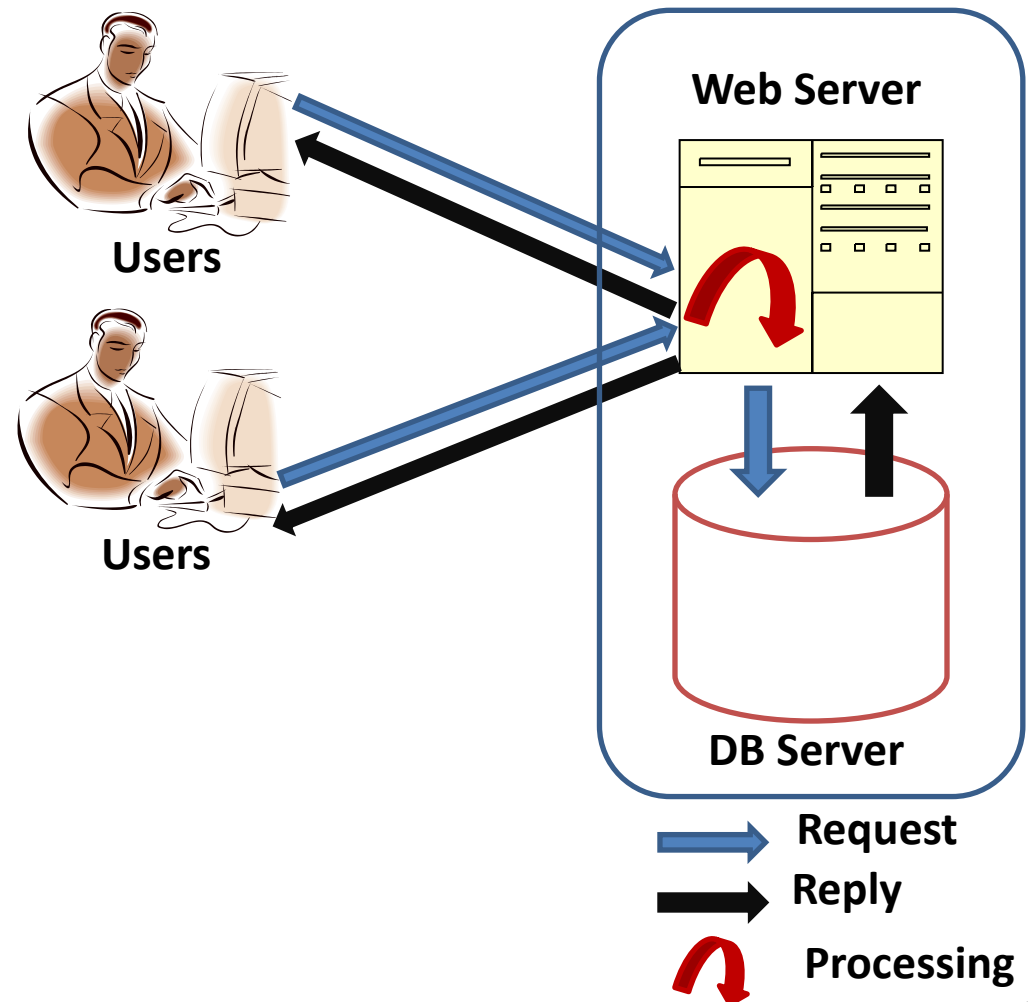
✗ Web app structure (Static One)





5. Overview on Web Development

✗ Web + Database app structure (Dynamic One)





6. Steps for web app development

- There are 9 steps – referring from SDLC.
 - **Problem definition**
 - Identifying the need for the web
 - **Requirement specification**
 - Highlighting the requirements (functionalities)
 - **System Analysis**
 - Describes the details of the functionalities, process, flow, and data.
 - **System Design**
 - Provides the view plan of the future system
 - **System Prototyping**
 - Develops minor core features of the intended system



- There are steps
 - **System Testing**
 - Test the functionalities of the prototype in the concerned users.
 - **System Implementation**
 - Escalate the system as a whole to all functionalities.
 - **Deployment**
 - As the web run online, you can deploy it to the remote server to be live (Go-live).
 - **Documentation/Training**
 - User guide/Manuals
 - System processes and features.

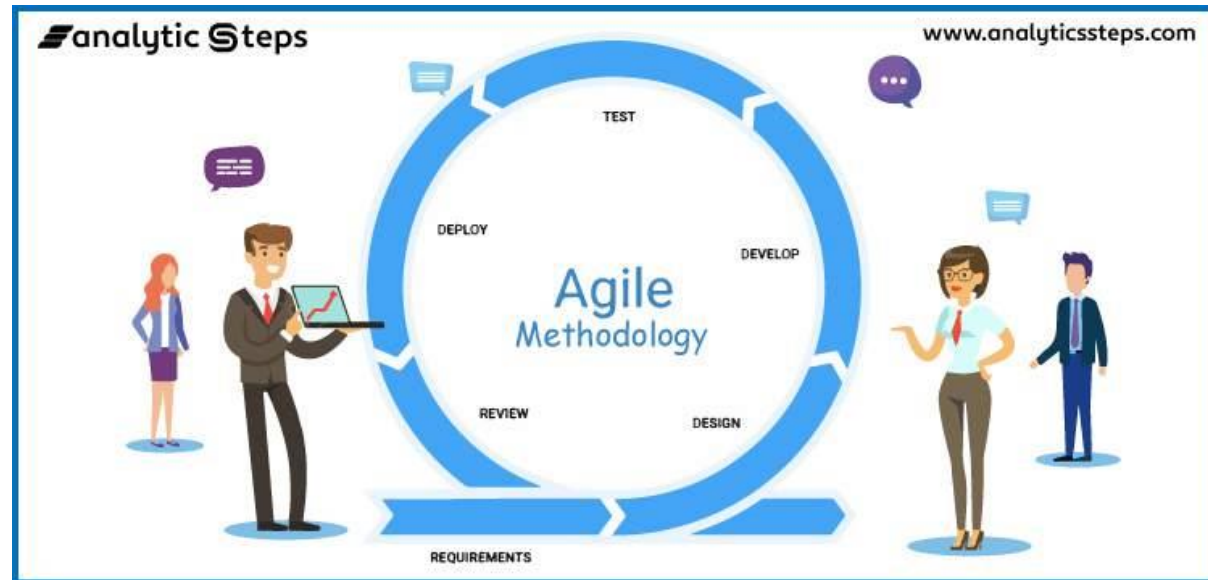
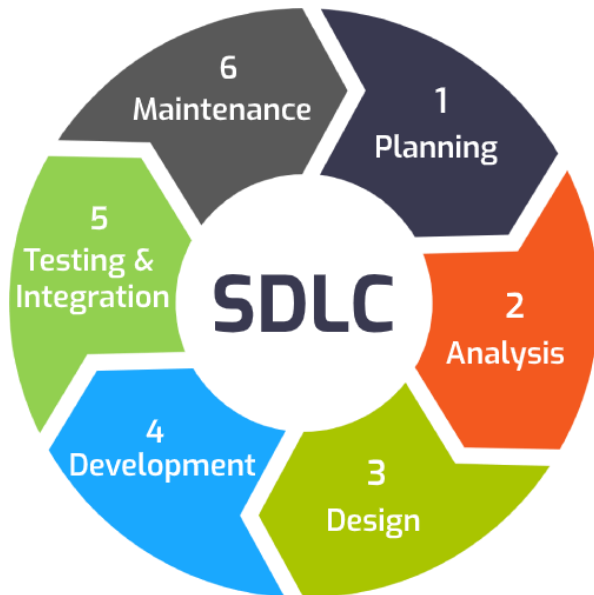


- Tools
 - **Methodology**
 - Refer to SDLC such as Agile, and DevOps.
 - **System Analysis**
 - Refer to UML tools (Draw.io.)
 - **System Design**
 - For Mockups (Figma, PS, Ms. PPT, ...)
 - **System Prototyping**
 - Any Editor: Notepad, Notepad++, Vs Code, Sublime,
 - **System Testing**
 - Depend on the technologies used, and features to be tested.



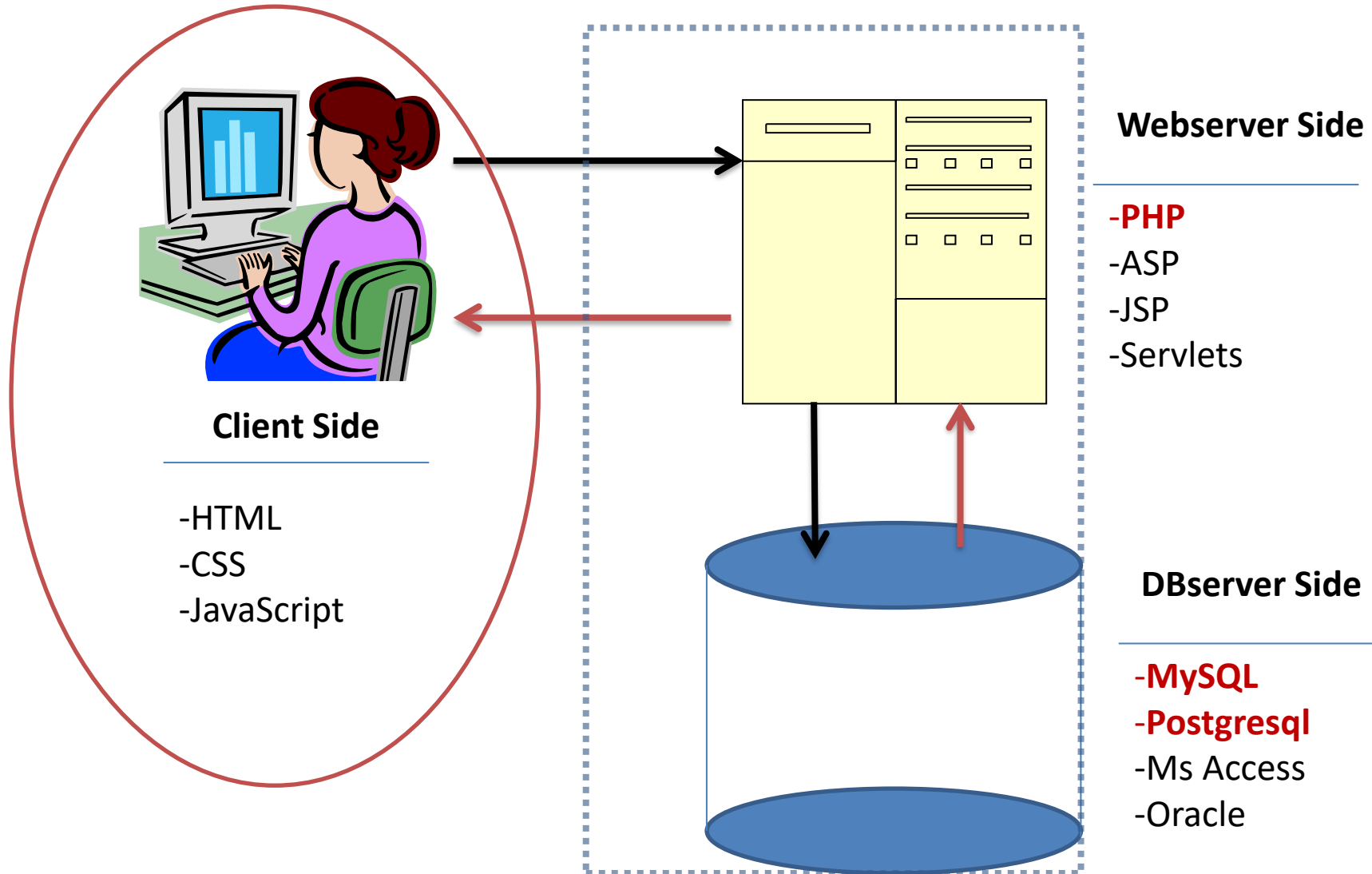
7. Tools for web sites design

- Tools
 - Methodology
 - Refer to SDLC such as Agile, DevOps, ...





8. Summary of Web Technologies





- Discuss the problem around which you think can be solved by the use of a Web Application. Two paragraphs. And the project title.
 - Identify the key users (beneficiaries) of that solution (Who shall benefit from the solution of the Web App). – Internal users.
 - Identify the key partners of this solution's ecosystem (who are the key personnel to understand and support those beneficiaries' solutions). - External users.
- => Make a hand-writing report of at least one paper with two sides.**